INCH-POUND

MIL-DTL-12883/10F
w/AMENDMENT 1
6 February 2018
SUPERSEDING
MIL-DTL-12883/10F
18 February 2003

DETAIL SPECIFICATION SHEET

SOCKETS AND ACCESSORIES FOR PLUG-IN ELECTRONIC COMPONENTS (ELECTRON TUBE, BOTTOM MOUNTING, SADDLE TYPE WITHOUT SHIELD BASE, 7 CONTACT, RADIAL)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and the: MIL-DTL-12883.

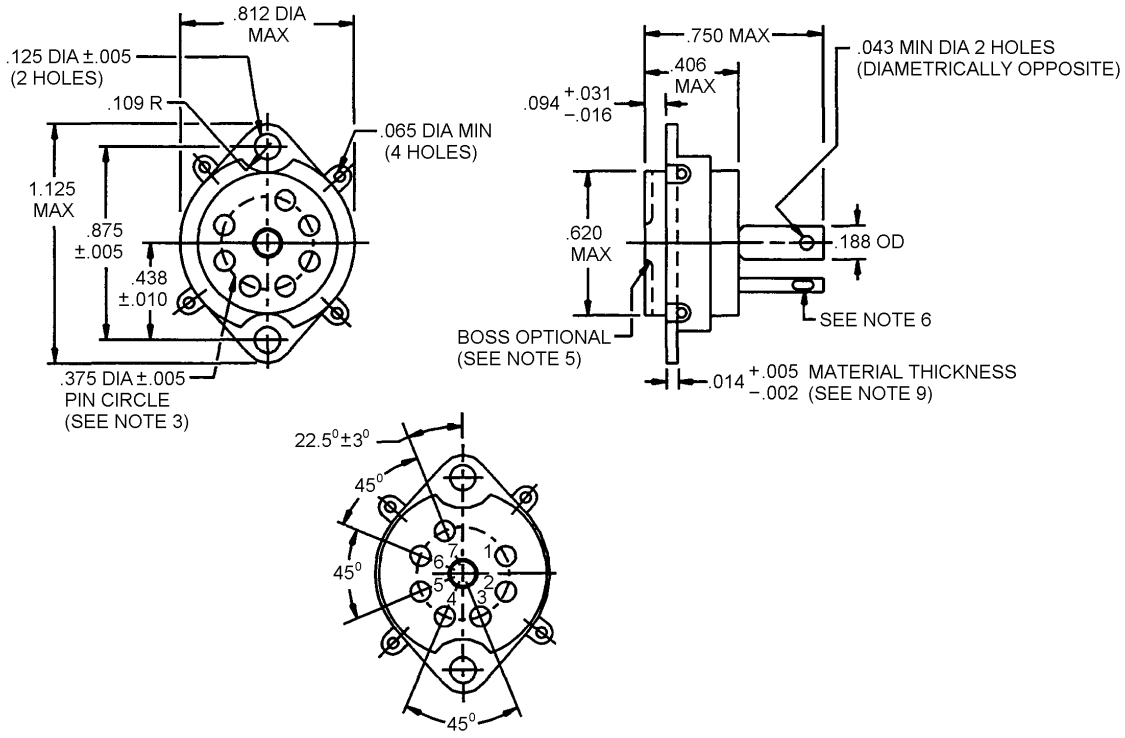


FIGURE 1. Socket configuration.

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Inches	mm	Inches	mm	Inches	mm	Inches	mm
.002	0.05	.031	0.79	.125	3.18	.620	15.75
.005	0.13	.043	1.09	.188	4.78	.750	19.05
.010	0.25	.065	1.65	.375	9.53	.812	20.62
.014	0.36	.094	2.39	.406	10.31	.875	22.23
.016	0.41	.109	2.77	.438	11.13	1.125	28.58

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ± .016 inch (0.41 mm) and ± ½º on angles.
4. Seven contact cavities equally spaced throughout 270º of arc, and each cavity located within ½º of true position, shall be established along the pin circle.
5. Each contact tab shall have either:
   a. 1 wire hole of .040 inch (1.02 mm) minimum width and .125 inch (3.18 mm) minimum length.
   b. Two holes of .040 inch (1.02 mm) minimum width and .075 inch (1.91 mm) minimum length.

   The contact tab hole, or holes shall lie on the longitudinal centerline of the contact tab within ± .008 inch (0.20 mm).
6. The design of bosses and the shape of barrier, when use, are optional. When present, barriers shall have a maximum height of .062 inch (1.57 mm).
7. Clearance between the cavity wall and the contact, with the contact in any position, shall be no greater than .018 inch (0.46 mm).
8. Marking shall be at any convenient visible location.
9. Material thickness of mounting flange shall be measured in an area where burring or dishing of the mounting hole is not present.

FIGURE 1. Socket configuration – Continued.

REQUIREMENTS:

Qualification: Not required.

First article testing: First article testing shall be as specified in the requirements of MIL-DTL-12883/10, tests shall be in accordance with MIL-DTL-12883.

Dimensions and configurations: See figure 1 and table I.

Insulating material: Ceramic, Diallyl Ortho-Phthalate in accordance with type SDG-F, GDI-30F, or mineral-filled Phenolic resin in accordance with ASTM D5948 type MFE, see table I.

Contact cavities: One shape only; D-shaped or circular, at option of the manufacturer.

Float: With a pin of .041 inch (1.04 mm) diameter and .271 inch (6.88 mm) minimum length fully inserted in a socket contact, the contact shall be capable of free movement (float) within the contact cavity.

Saddle: The method of attaching the saddle to the insulator body, and the shape of the saddle where attachment is effected, are optional.

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Electrical:

Insulation resistance: 1,000 megohms minimum. Test pin diameter: .040 ± .001 inch (1.02 ± .03 mm).

Dielectric withstanding voltage:

Sea level: Test voltage: 2,000 volts root mean square (ms). Test pin diameter .040 ± .001 inch (1.02 ± .03 mm).

High altitude: Test voltage: 660 volts ms. Test pin diameter: .040 ± .001 inch (1.02 ± .03 mm).

Contact resistance:

Average for all contacts: 0.015 ohm maximum.

Individual contacts: 0.03 ohm maximum.

Continuity test circuit: With the header test-gage(s) inserted into the socket(s) under test, the pins of the test-gage(s) and the contacts of the socket(s) under test, shall result in a series circuit. The header shall be the one used for electron tube 6AU6WB.

Mechanical:

Insertion and withdrawal force:

Initial insertion force: 15 pounds (67 newton) maximum.

Average insertion force: 12 pounds (53 newton) maximum.

Initial withdrawal force: 4 pounds (18 newton) minimum, 12 pounds (53 newton) maximum.

Vibration: The test gage shall be an approved electron-tube type 6AU6WB or equivalent, in accordance MIL-PRF-1 test method 1031.

High-frequency vibration:

Conditions Rp = 2,000 ohms
Acceptance level = 6.5 (This specification sheet utilizes in accept on zero defect sampling plan in accordance with MIL-PRF-1, table III).
Symbol = Ep
Limits: max = 300
Unit = mV ac

Vibration-fatigue:
Acceptance level = 6.5 (This specification sheet utilizes in accept on zero defect sampling plan in accordance with MIL-PRF-1, table III).

Low-frequency vibration:

Symbol = Ep

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Limits: max = 300
Unit = mV ac

Durability: After the durability test, the withdrawal force shall be 2 pounds (9 newton) minimum.

Static load: 40 pounds (178 newton).

The part shall be designed to operate at the following conditions:

Environmental:

Operating temperature: See table I.

TABLE I. Dash number and characteristics.

Dash number	Operating temperature ºC	Insulating material	Old type designator 1/
-01	200	Ceramic	TS102C02 (2-way)
-02	100	SDG-F, GDI-30F	TS102P02 (1-way)
-03	200	MFE	TS102P02 (2-way)

1/ 1-way replacement: The new item (designated by military part number) will replace the old item (formerly designated by type designator); however, the old item will not replace the new item.
2-way replacement: The new items are interchangeable; i.e., the new item will replace the old item, and the old item will replace the new item.

Ratings (absolute maximum):

Voltage:

Sea level: 500 volts rms.
50,000 feet (15.24 km): 225 volts rms.

Current: 5 amperes.

Operating temperature: See table I.

Test gage details: See table II and MIL-DTL-12883 (see appendix).

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TABLE II. Test gage details. 1/ 2/

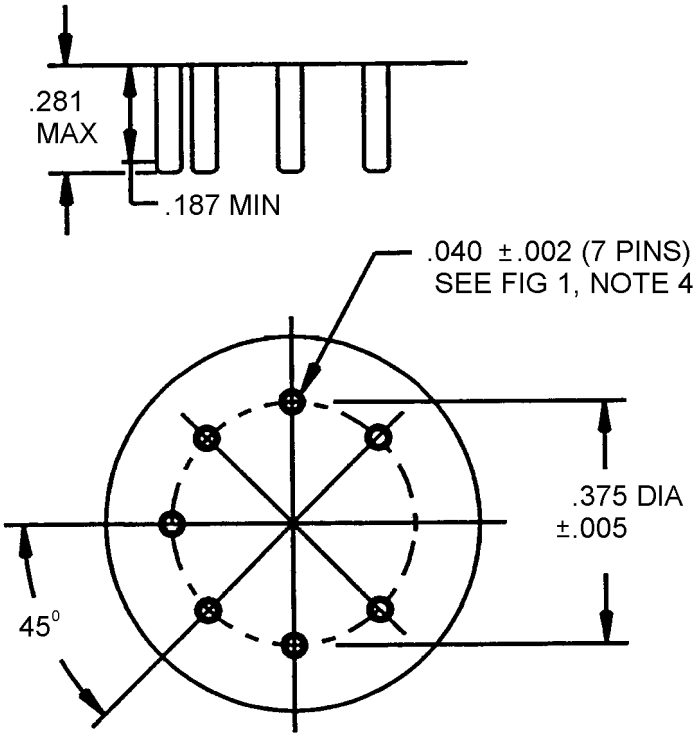
Inspection	A Pin length (mm)	B Pin dia (mm)	M Test-end dia (mm)	C Pin-circle dia (mm)	N Probe-end dia (mm)
Insertion and withdrawal force	.276 ±.005 (7.01 ±0.13)	.0390 ±.0001 (0.991 ±0.003)	---	.3750 (9.525) basic	---
Contact resistance	.276 ±.005 (7.01 ±0.13)	.0390 ±.0001 (0.991 ±0.003)	---	.3750 (9.525) basic	---
Contact retention	.276 ±.005 (7.01 ±0.13)	---	.0390 ±.0001 (0.991 ±0.003)	---	.0410 ±.0001 (1.041 ±0.003)
Durability	.276 ±.005 (7.01 ±0.13)	.0410 ±.0001 (1.041 ±0.003)	---	.3750 (9.525) basic	---

Inspection	D Gage dia max (mm)	No. of pins	Total weight of gage ounces (grams) (±5%)
Insertion and withdrawal force	11/16 (17.46)	7	---
Contact resistance	11/16 (17.46)	7	---
Contact retention	---	---	3 (85.05)
Durability	11/16 (17.46)	7	---

1/ Dimensions are in Inches.
2/ Metric equivalents are given for general information only.

Mating-base dimensions: Sockets shall accommodate plug-in components having mating-base dimensions as shown on figure 2.

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Inches	mm
.002	0.05
.005	0.13
.040	1.02
.187	4.75
.281	7.14
.375	9.53

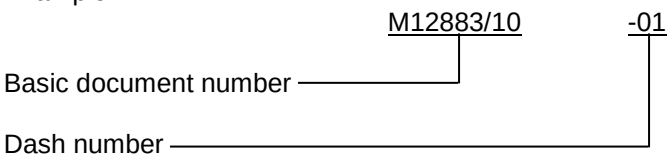
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FIGURE 2. Mating-base dimensions.

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Part or Identifying Number (PIN): The PIN shall consist of the basic number of this specification sheet and the dash number from table I.

Example:



The Government PIN, specified in table III, supersedes the following commercial PINs.

TABLE III. Supersession and cross reference data.

Active Government PIN	Superseded PIN
	CAGE 72825
M12883/10-01	9726-153-03

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Reference documents. In addition to MIL-DTL-12883, this document references the following:

MIL-PRF-1
ASTM-D5948

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA – CC

Preparing activity:
DLA – CC

(Project 5935-2018-024)

Review activities:
Army – AR, AT, CR4, MI
Navy – AS, CG, MC, OS, SH
Air Force – 70, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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